

DATA/CS 172: Python for Data Analysis

Getting Started with Python

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UNIVERSITY
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HILO

Today

- Admin Wrap up
- Getting started with Python
- Variables & Values
- Printing

Note about slides

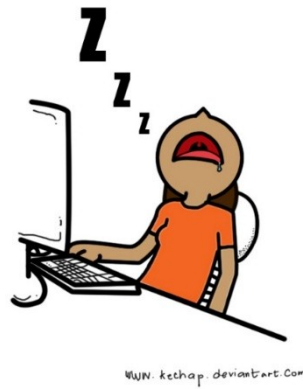
- Slides posted after class
- Intended as memory-joggers, not comprehensive notes!
 - Good idea to take your own notes as well to aid comprehension and to help study

Cheating/Academic integrity

- Copying another student's solution?
 - **Cheating**
- Googling for a solution to the assignment?
 - **Cheating**
- Googling how to print out a number?
 - **Not cheating – short, simple, just a memory issue**
- Asking AI to write a chunk of code for you?
 - **Cheating – need to learn how to code on your own before using these tools (also functionality, style, ethical issues)**
- Copying an example from the book?
 - **Not cheating**
- Showing/Sending someone else your solution, telling them to use it as a reference only?
 - **Cheating**
- Talking at a high level with another student about a general approach to solving the problem?
 - **Not cheating**
- Having a friend who is a really good programmer write the code for you on a whiteboard, while you type it in?
 - **Cheating**
- Giving another student who is stuck a couple high-level hints?
 - **Not cheating**
- Working together with another student – talking through numerous details of how to write the assignment?
 - **Cheating – Programs are meant to be YOUR INDEPENDENT WORK**

Getting started with Python

Start up IDLE



- IDLE stands for Integrated Development and Learning Environment.
- Basically a fancy text editor
 - Colors highlight useful parts
 - Won't get in your way
- If you hate IDLE, could use other environments like Notepad++, Pycharm, Xcode, vim, emacs, etc.
 - Mostly need a way to edit the text
 - I think Pycharm is very nice once you get to large projects with many different scripts
 - Not usually the case in this class
 - Can't give specific help on the features of these all of these programs
 - For now, follow along in IDLE

When you start IDLE...

- It's running python in **interactive mode** ← **Important term!**
- As soon as you type a command and hit "enter", the command is run and the output is shown
- But, you are typing real Python **code!** ← **Important term!**
 - Aka text that Python can interpret and run
- Note for C++ programmers: This wouldn't really be possible in C++, because C++ is a compiled language where code must be compiled before it is run.
 - Python is designed as an interpreted language, which adds flexibility at the cost of speed.



Demo

Demo

- "hello world"
- $5 * 3$
- $a = 6$
- $a * 2$
- $result = a * 2$
- result

What are these weird names called (a, b, etc?)

- Any guesses?
 - Hint: Very similar to x, y, etc. in math
 - Right! **Variables** ← Important term!
- What do we call those things that we put in variables?
 - **Values** (same as math) ← Important term!
- When we write a=6, that puts the value 6 in the variable a
 - Called **assigning**... ← Important term!
 - Variables can be assigned multiple times (demo)
 - Overwrite each other
 - First time is called **defining** or **initializing** the variable ← Important terms!



HELLO
my name is

?

Note: Variable Names

- Which is a better name?
 1. x
 2. number
 3. dollars
 4. costOfEquipment
 5. costOfEquipmentInDollarsIncludingUtilitiesAndTax
- More descriptive names better (to an extent)!
- Cannot combine words with space in variable names
 - can_use_underscores
 - orCamelCase (preferred by many)

Scripts



- Right now we're using Python kind of like a calculator
- What if I ask you to re-do the calculations, but change a few numbers?
- As a data scientist, what if we collect more data and want to throw it in to our analysis?
 - Certainly wouldn't want to retype all the commands!
- Want to create a big list of commands that run all at once
 - Called a **script** or a **program** ← Important terms!

To start a script, click File→New

- Type your script
- Save your script File→Save
- Then to run the script, hit Run→Run Module
 - Or just hit F5
- demo

```
1      EXTERIOR - GARDEN UNDER JULIET'S WINDOW - NIGHT      1
Rosenkranz and Guildenstern climb the wall and enter the
garden.

      ROSENKRANZ
      There is a light in her window.

      GUILDENSTERN
      (total ham)
      Ah ha! That must be the east and
      Juliet is the sun.

      ROSENKRANZ
      You're weird!

      GUILDENSTERN
      Hark, she comes!

      As if in a dream, Juliet Capulet emerges from her bedroom
      onto her balcony. She is picking the peddles off a flower and
      humming a song.

      ROSENKRANZ
      It is my lady.

      GUILDENSTERN
      Oh, it is my love.

      JULIET
      Ahhhh! Me!

      ROSENKRANZ
      She speaks.

      He moves forward then hesitates.

      ROSENKRANZ (CONT'D)
      But I am too bold.

      GUILDENSTERN
      Yes, perhaps it is not to me that
      she speaks.

      JULIET
      Romeo, Romeo. Wherefore art thou
      Romeo?

      ROSENKRANZ
      Alas, that is not I.

      GUILDENSTERN
      Nor I. Crushed.
```

Printing



- In computer science, **printing** usually refers to putting text on the screen, not on paper ← Important term!
- In the interactive window, it kind of prints automatically
 - Not so when writing a script! (demo)
- To put output on the screen, have to use `print()`
 - `print("hello world")`
 - Quotes mean print the text exactly (instead of treating it as a variable)

Printing with variables

```
totalHomes = 6 * 7
```

```
print("The number of total homes is:")
```

- What now?
 - Idea 1:

```
print("The number of total homes is:")
print(totalHomes)
```

 - Have to print on separate line which is annoying
 - `print ("The number of total homes is:", totalHomes)`
 - *For CS majors: print is a function with any number of arguments*
- A (more advanced) way to do this is f-strings
 - `Print(f"There are {totalHomes} homes")`
 - F in front, variables enclosed in curly brackets
 - Can be a little less typing

Demo of print with vars

Comments



- Lines that start with a `#` are treated as a **comment** ← Important term!
 - Means you can write anything, no effect on how program runs
 - Useful to write notes about what the program is doing

Whew! That's enough for now...

- Time to get going on the rest of the first lab!
- Next time: Getting started with data